

The Multilayer Perceptron and StockMixer Ensemble

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Abstract

This document describes an ensemble of two cross sectional models for monthly portfolio construction, a plain multilayer perceptron and a StockMixer model. Both are trained on the same rolling window under a maximum Sharpe ratio regression objective, and their monthly weights are blended equally and renormalised. We develop both constituents and the shared pipeline in full, present the blending rule, explain the diversification argument that motivates it, and report the realised performance.

1 Overview and motivation

Two networks may share an objective, a data pipeline, and a training protocol, and yet differ in how they couple the cross section, so that they make partially independent errors. A combination of their books can then have a lower variance than either alone without sacrificing the mean, which raises the Sharpe ratio. The two constituents here are chosen for exactly this reason, since one couples the cross section through a fixed pooled average and the other through a learned exchange with a market state. The purpose of the ensemble is therefore diversification across model specifications rather than any new selection signal.

2 The shared pipeline

The raw inputs are a panel of monthly characteristics and a list of candidate features, and the target is the excess return of a security over the following month. A characteristic is retained if its coverage is at least ninety percent over the pre test period, which makes the selection causal and algorithmic. For each month every retained characteristic is transformed to its within month percentile rank, missing entries are set to the median before centring so that the inputs lie in the interval $[-0.5, 0.5]$, and securities missing more than one third of the characteristics are dropped. Each constituent is fitted under the maximum Sharpe ratio regression loss $(1 - \sum_i w_i r_i)^2$, which drives the realised managed return towards unity in every month, on a rolling window of at most sixty months, with a fifty epoch cold start, ten epoch warm starts thereafter, three seeds, and seed averaging of the weights normalised to unit gross exposure.

3 The two constituents

The ensemble is standalone only if both architectures are specified. The first is a plain perceptron. Each security is encoded to an embedding z_i , a pooled context $c = n^{-1} \sum_i z_i$ is formed as the average embedding across the month, and the score is read from a head on the concatenation $[z_i, c]$, so that a security’s score depends on the composition of its cross section through a single fixed average.

The second is a StockMixer. It first mixes information across the characteristics of each security through a residual perceptron applied to the input vector, and then couples the cross section through a learned exchange. A stock to market step pools the security tokens into a market state $m = n^{-1} \sum_i \text{to_market}(z_i)$, and a market to stock step broadcasts that state back and updates each token through $z_i \leftarrow z_i + \text{from_market}([z_i, m])$, after which the score is read from a head. The two models therefore differ precisely in how the cross section is summarised and reinjected, which is the source of the diversification the ensemble exploits.

4 The blending rule

For a given month let w^a and w^b denote the weight vectors of the two constituents, aligned on the union of the securities they hold and filled with zero where a security is absent from one of them. The blended weight of security i is the equal average

$$w_i = \frac{1}{2}w_i^a + \frac{1}{2}w_i^b, \quad (1)$$

and the blended vector is renormalised within the month to unit gross exposure,

$$\tilde{w}_i = \frac{w_i}{\sum_j |w_j|}. \quad (2)$$

The weights are equal and fixed rather than estimated, which avoids introducing a further layer of parameters that would itself require validation and would be prone to overfitting. We can observe that because each constituent is already normalised, the equal average gives the two models comparable influence over the blended book before the final renormalisation restores unit gross exposure.

5 Evaluation and concluding remarks

On the evaluation window the ensemble attains a Sharpe ratio of 2.405 in the volatility scaled form and 2.438 in the unscaled form. The plain perceptron alone attains 2.484 and the StockMixer alone attains 2.280 on this panel, so the ensemble lies between the two constituents and exceeds their simple average, which is consistent with a diversification benefit from combining partially independent errors. We conclude that the ensemble delivers a stable book whose performance is close to that of the stronger constituent while drawing on both, and that the equal weighting is an appropriately conservative combination rule given that the constituent Sharpe ratios are similar and their relative ordering out of sample cannot be known in advance. Within the limitations of the present analysis, the ensemble is a robust choice when the aim is stability rather than the maximisation of the in sample Sharpe ratio.